CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm {Pharmaceutical Technology (PT) and [Pharmaceutics (Drug Regulatory Affairs)]}

University Theory Examination December-2016

PHCC001 Basic Analytical Techniques

Date: 12/12/2016, Monday Time:10:00 a.m. to 01:00 p.m. Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1	**(A)**	Differentiate between TGA and DTA. Draw a schematic diagram of calorimeter.	5
	(B)	Discuss about SDS gel electrophoresis.	5
Q.2	**(A)**	How will you perform acid and base degradation study? How the mass balance is calculated at the end? Explain it giving suitable example.	5
	(B)	Enlist methods for identification of impurities. Explain use of PDA detector for the same.	5
		OR	
Q.2	**(A)**	Explain the instrumentation of XRD.	5
	(B)	Give importance of method validation and discuss estimation of any two parameters as per ICH guidelines.	5
Q.3		**Answer the following (Any three)**	**15**
	(A)	Explain the sources, record and storage for reference standards.	
	(B)	Discuss different components of Scanning electron microscopy.	
	(C)	Discuss applications of DSC.	
	(D)	What is LLOQ and LLOD? How accuracy study should be performed for bioanalytical method validation?	

SECTION-II

Q.4	**(A)**	Discuss factors affecting absorption position (λmax) in UV spectroscopy giving suitable examples.	5
	(B)	Write about sample handling techniques in IR spectroscopy.	5
Q.5		Explain derivatization in HPLC and HPTLC.	10
		OR	
Q.5		Classify chromatographic technique. Discuss instrumentation for flash chromatography.	10
Q.6		**Answer the following (Any three)**	**15**
	(A)	Explain Factor affecting Chemical Shift in detailed.	
	(B)	Explain principle of Mass spectroscopy. Classify ionizing techniques used in Mass spectroscopy. Describe any one in detail.	
	(C)	Mc Lafferty and Diel's Alder rearrangement	
	(D)	Explain instrumentation for NMR.	
